

Chapter 5 – Autoregressive and Moving Average Models

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1 The ACF and the PACF

1. In Chapter 4, we see that the correlogram given by the acf of a time series does give us important information about the series, whether there is a trend or if there is seasonality.
2. However, in order to find out more accurately if, say the series x_t at any time t is significantly correlated with the observation x_{t-k} at k lags before it; or whether it appears to be correlated with x_{t-k} because it is correlated with x_{t-1} at 1 lag and the apparent correlation with x_{t-k} only comes about because x_{t-k} is correlated with x_{t-1} and other observations in between.
3. When we look at the partial correlogram, the *partial autocorrelation function* (pacf) is calculated instead.
4. The partial autocorrelation at lag k is the correlation between x_t and x_{t-k} that results after removing all the correlations due to intermediate lags.
5. We will not be concerned with the exact mathematics with how this removal of the correlations due to the intermediate lags are done. (This involves some theory of projections in orthogonal spaces which is beyond the scope of this course). However, you should be able to interpret the partial correlogram output produced by **R** .
6. In order to see the difference between the correlogram and the partial correlogram, we will be looking at simulated data first.

7. We will simulate a stationary autoregressive $AR(1)$ process. Recall that this takes the form

$$x_t = \alpha x_{t-1} + w_t,$$

where $\{w_t\}$ is white noise process and $|\alpha| < 1$.

8. **Example 1.** We will try one round of simulation with 100 periods with $\alpha = 0.5$. So the model is

$$x_t = 0.5x_{t-1} + w_t,$$

and each w_t is normally distributed $N(0, 1)$.

```
> set.seed(1)
> x <- w <- rnorm(100)
> for (t in 2:100) x[t] <- 0.5 * x[t-1] + w[t]
> plot(x, type = "l")
> acf(x)
> pacf(x)
```

In the first line of the code, we set the random seed. In the second line, 100 values of x_t are set to be the same as that of the randomly generated w_t that are independently and identically distributed $N(0, 1)$ random variables. In the third line, the for loop replaces each x_t for $t = 2, \dots, 100$ as the weighted sum (weighted by $\alpha = 0.5$) of the previous term x_{t-1} and the current random white noise w_t . Finally we have the 3 plots: the plot of the time series x in Figure 1, the correlogram in Figure 2 and the correlogram in Figure 3. Note that your plots are going to look different since you are going to get different simulated values of w_t and x_t .

Figure 1: Plot of Simulated $AR(1)$ Process

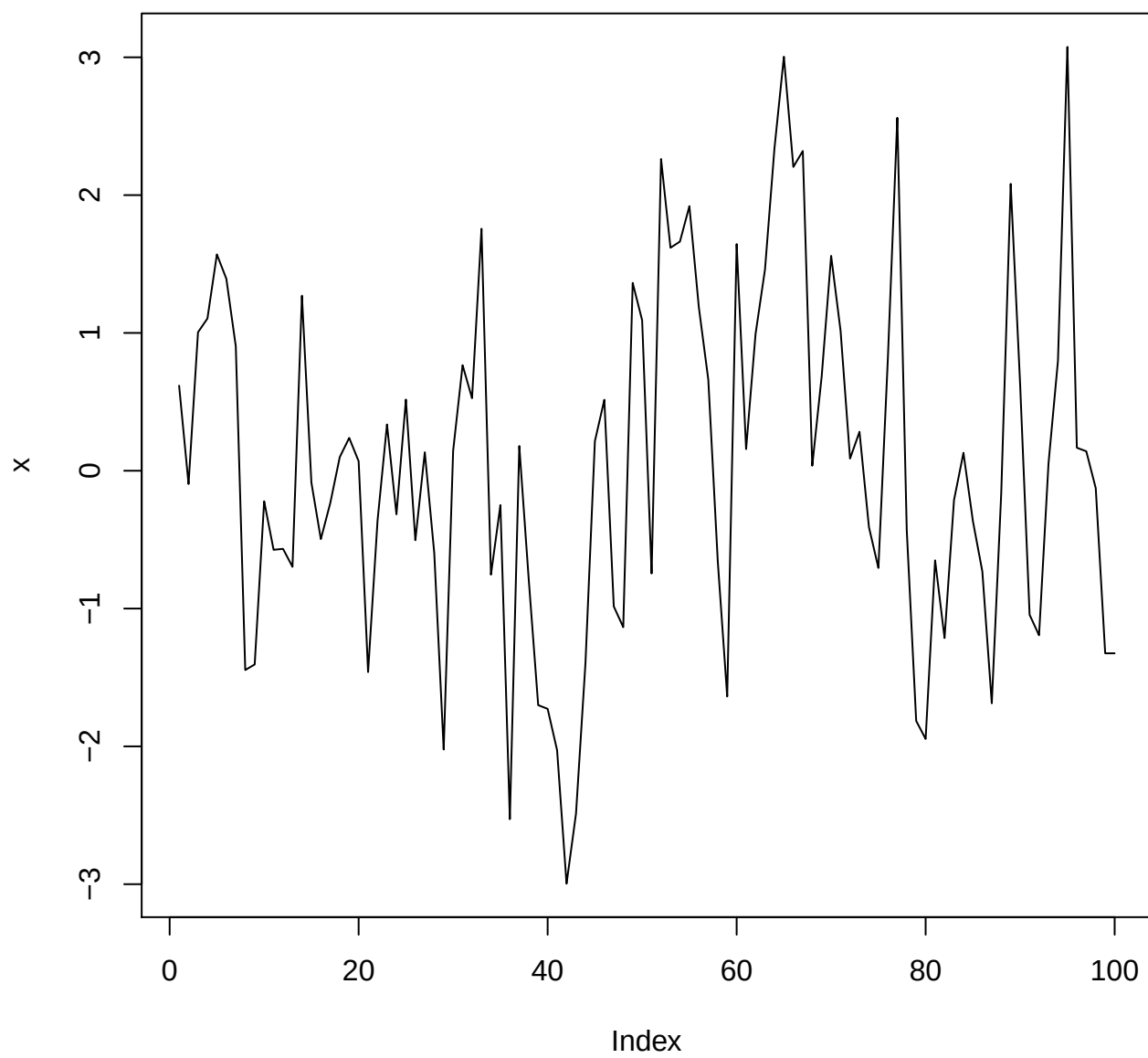


Figure 2: Correlogram of Simulated $AR(1)$ Process

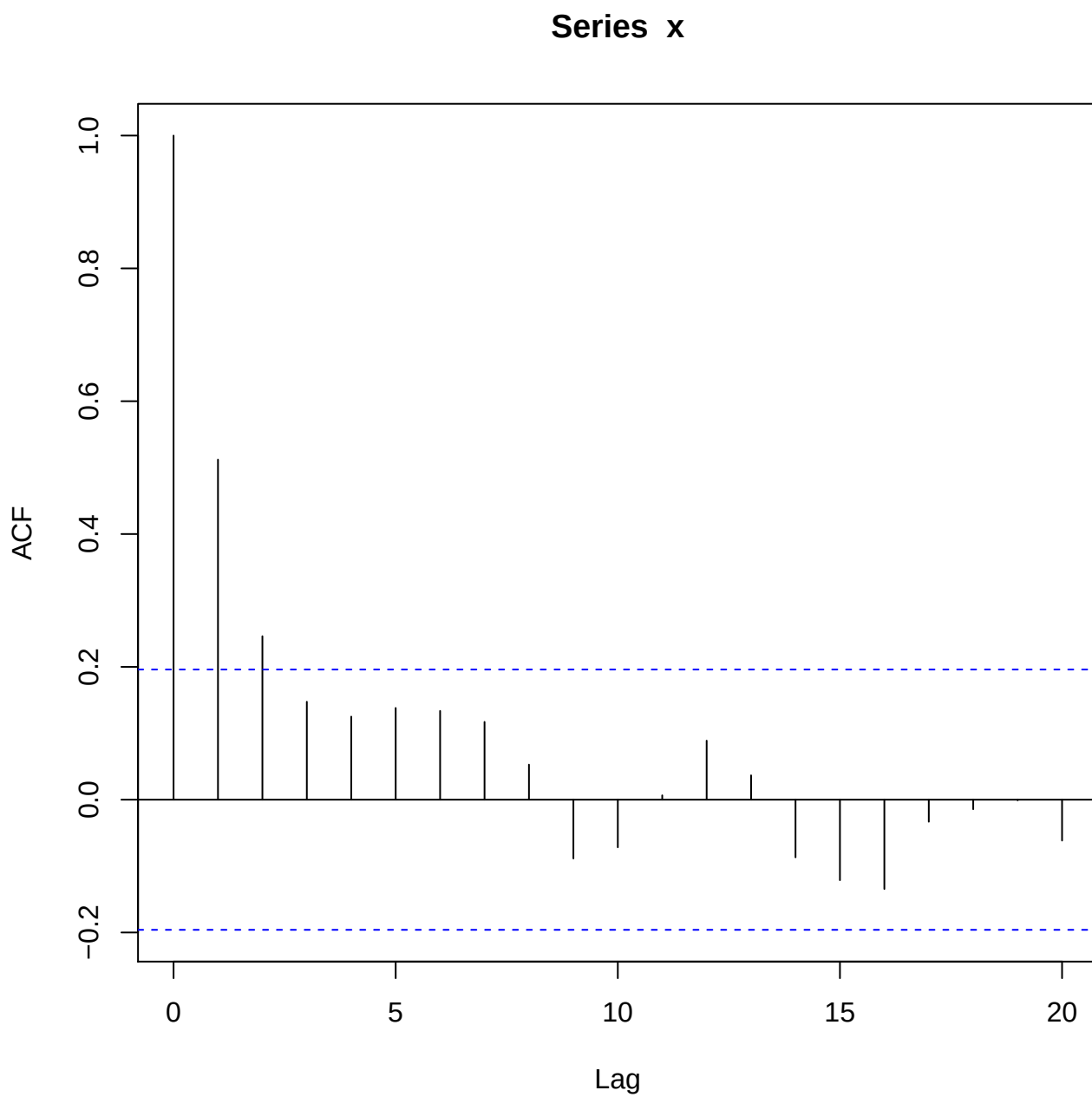
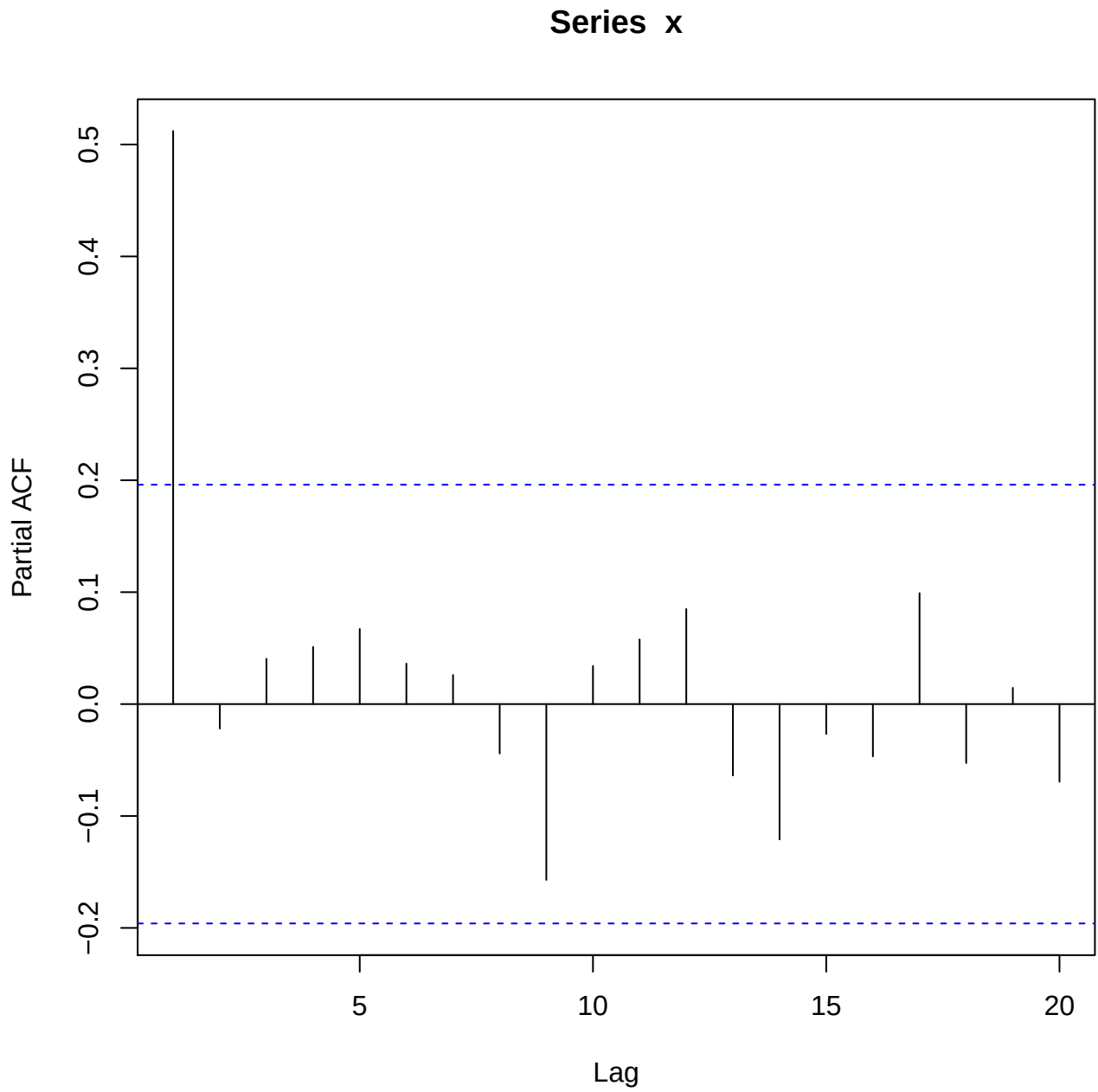


Figure 3: Partial Correlogram of Simulated $AR(1)$ Process



In Figure 2, the correlogram shows that the simulated series is correlated at lags 1 and lags 2. However, we do not know whether the correlated with lag 2 is entirely due to the correlation with lag 1 in the first place. In Figure 3, when we look at the partial correlogram, then we know that there is only correlation at lag 1 only, which is to be expected since this is an $AR(1)$ process.

If we use the command `pacf(x)[ ]`, this lists out the values of the partial correlation at all the lags. In this case the first value is 0.512 and the rest fall within the blue dotted lines, which mean that they are insignificant. So this tells us that for this particular simulated $AR(1)$ series, the estimate of α is 0.512 which is quite close to the theoretical value of the 0.5 that we have chosen. The simulated series also shows partial correlation at other lags but these are insignificant. These results are to be expected since we deliberately set out to simulate an $AR(1)$ model with $\alpha = 0.5$ in the first place. You can try other values of α including negative and positive values and values of $|\alpha| > 1$.

9. It can be shown that for an autoregressive $AR(1)$ process $\{x_t\}$, the mean is always

$$\mu_k = E[x_k] = 0,$$

the covariance at lag k is

$$\gamma_k = \text{Cov}[x_t, x_{t+k}] = \frac{\alpha^k \sigma^2}{(1 - \alpha^2)},$$

and the autocorrelation at lag k is

$$\rho_k = \alpha^k.$$

Here σ^2 is the variance of the white noise.

10. **Example 2.** Suppose we consider the $AR(1)$ model

$$y_t = -1.01y_{t-1} + w_t.$$

Here $\alpha = -1.01$ and thus $|\alpha| > 1$. The plot in Figure 4 for a simulation of the model shows why it is not stable. Similar instabilities for higher

order autoregressive models (see Chapter 4, Section 5) if at least one root of the characteristic equation for the polynomial in B satisfies $|\alpha| > 1$.

2 Autoregressive Models

1. Recall that an autoregressive model $AR(p)$ of order p takes the form

$$x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} + \cdots + \alpha_p x_{t-p} + w_t.$$

2. We will now work through a real data set to see if we could fit an $AR(p)$ model to it.
3. This data set `global.txt` consists of global average temperature deviations in Celcius from monthly means, recorded from January 1856 to December 2005.
4. First we read the data in.

```
> Global.dat <- scan("D:/MATH 5045/Data/global.txt")
Read 1800 items
> Global.ts <- ts(Global.dat, st=c(1856,1), end = c(2005,12), fr = 12)
```

The `ts( )` command interprets the data as a time series, where the frequency is 12 within each year, so that the data is interpreted as monthly data. The parameters for `ts( )` indicated the 1st month of 1856 and that of end, indicate the last month of 2005.

5. A plot of the temperature series is found in Figure 5.

Figure 4: Plot of Simulated Unstable $AR(1)$ Process

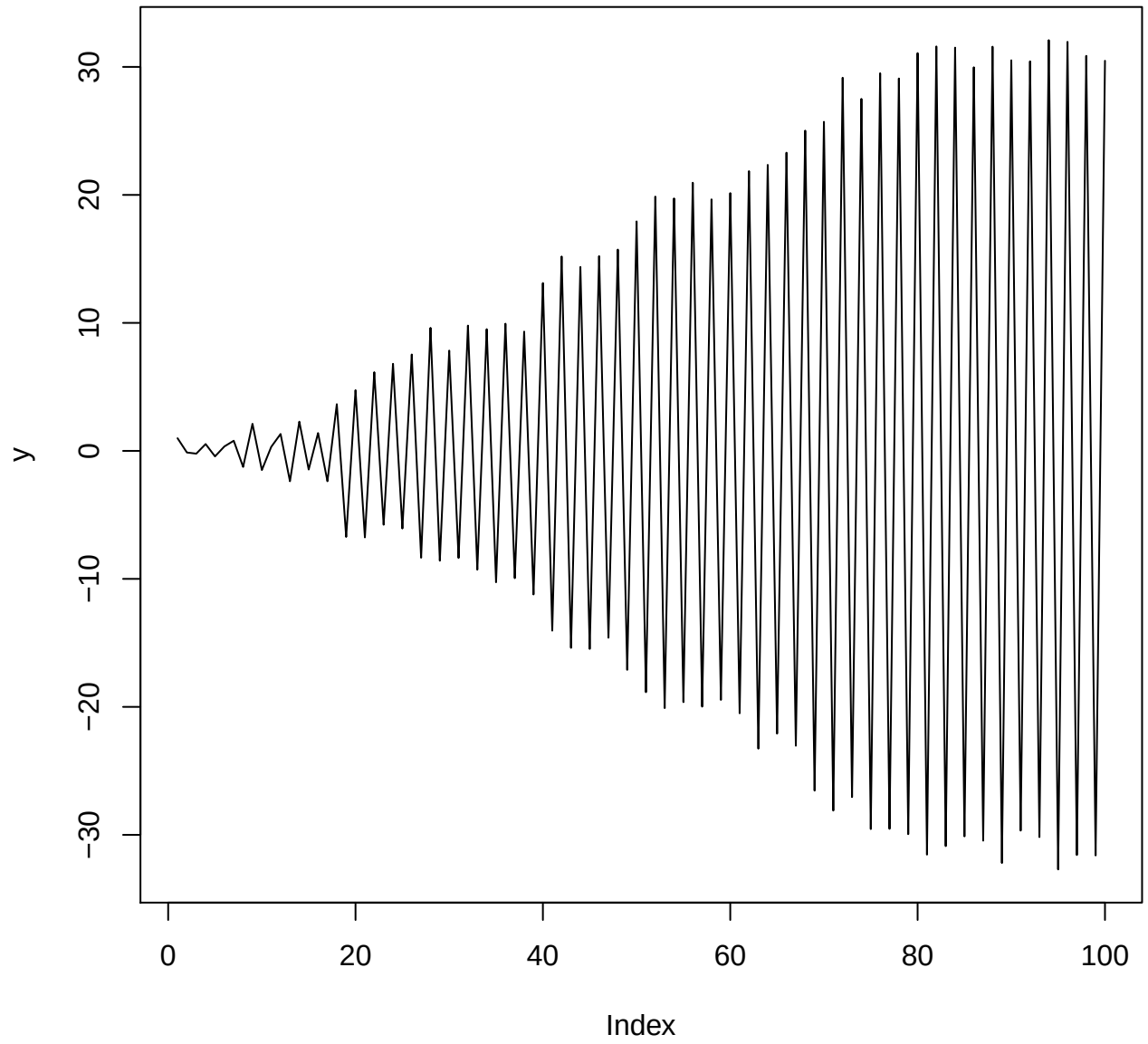
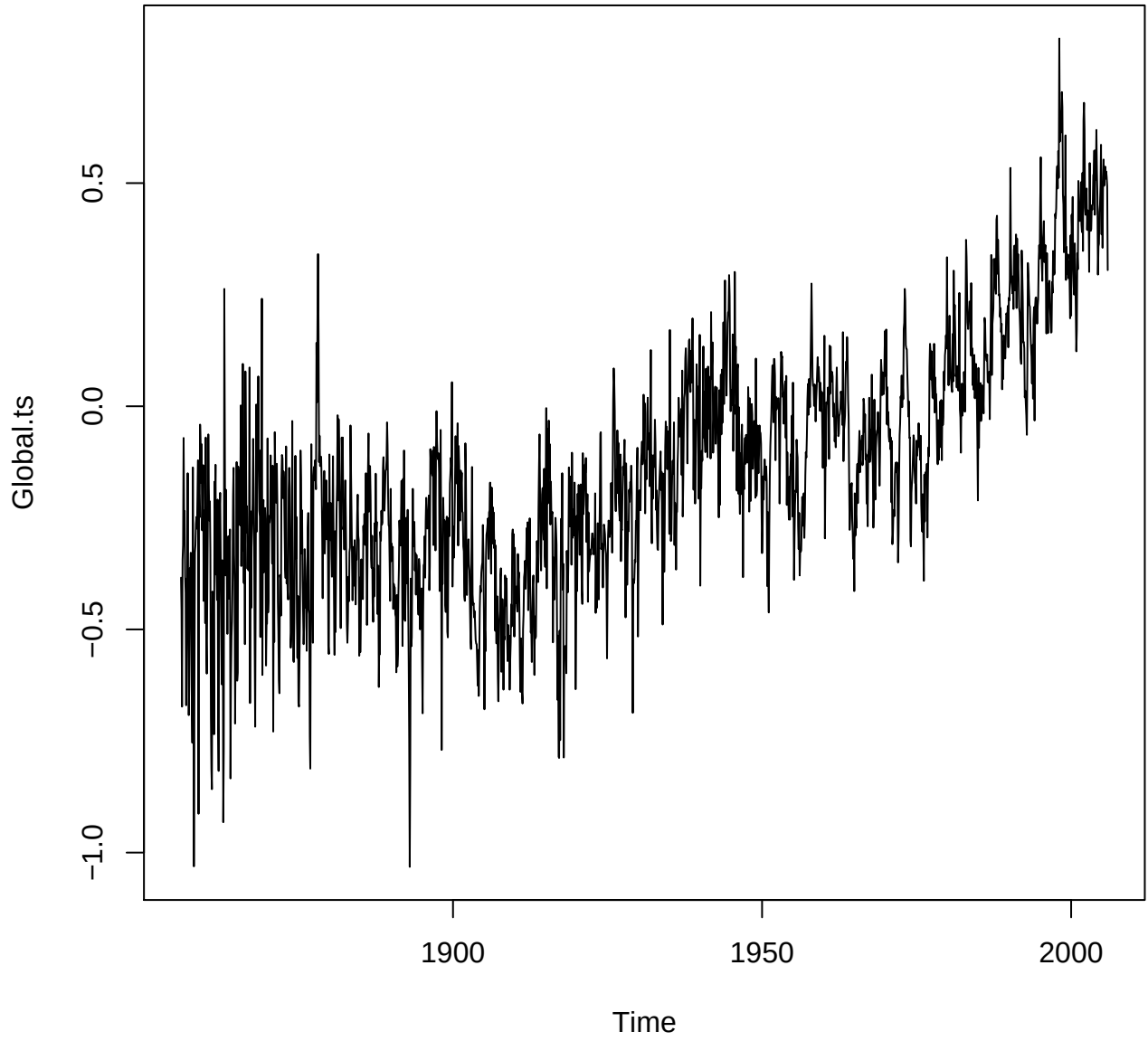


Figure 5: Monthly Temperature Deviations from 1865 to 2005



6. It looks like there might be a trend but it is not clear whether there is a trend or just randomness that happens to drive the average temperature deviations upwards from say around 1970 onwards.
7. In order to look clearly at the annual trends rather than the monthly seasonal changes, we can aggregate the data into its annual mean temperature deviations from each year, rather than the monthly mean temperature deviations.

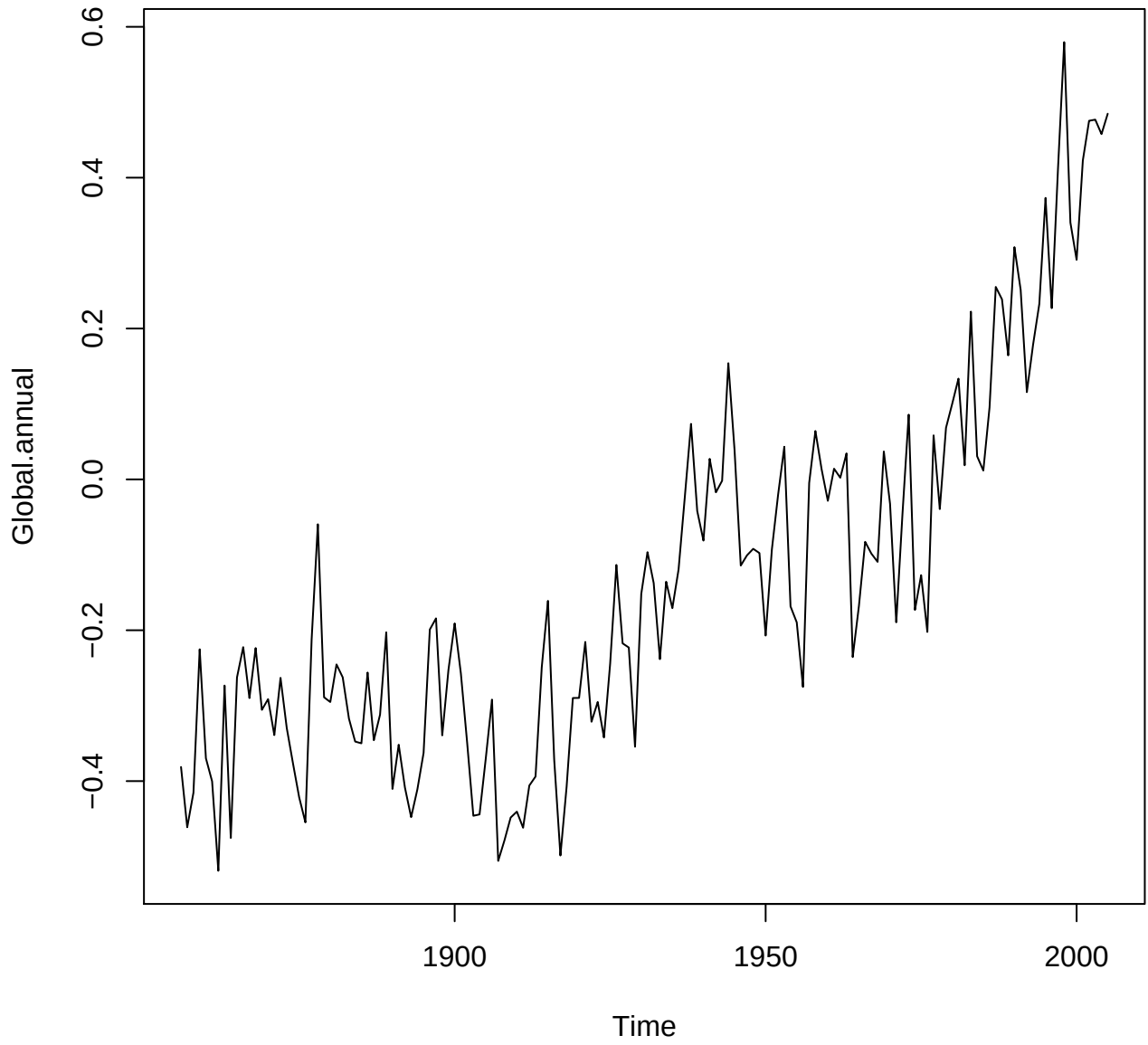
```
> Global.annual <- aggregate(Global.ts, FUN = mean)
```

The plot is shown in Figure 6.

8. There is a command `ar( )` in **R** that fits an autoregressive model to the data and figures out the order of the lags that are significant in the model. However, this only works well with a stationary series (or an assumed stationary) series with mean close to zero.
9. Thus we check its mean μ_x and then remove its mean to form another time series

$$y_t = x_t - \mu_x.$$

Figure 6: Yearly Temperature Deviations from 1865 to 2005



10. In **R** this is how we remove the mean and run the `ar( )` function on y_t .

```
> mean(Global.annual)
[1] -0.1382628
> Global.annual.mean.adjusted <- Global.annual - mean(Global.annual)
> Global.ar <- ar(Global.annual.mean.adjusted, method = "mle")
Warning message:
In arima0(x, order = c(i, 0L, 0L), include.mean = demean) :
  possible convergence problem: optim gave code = 1
> Global.ar

Call:
ar(x = Global.annual.mean.adjusted, method = "mle")

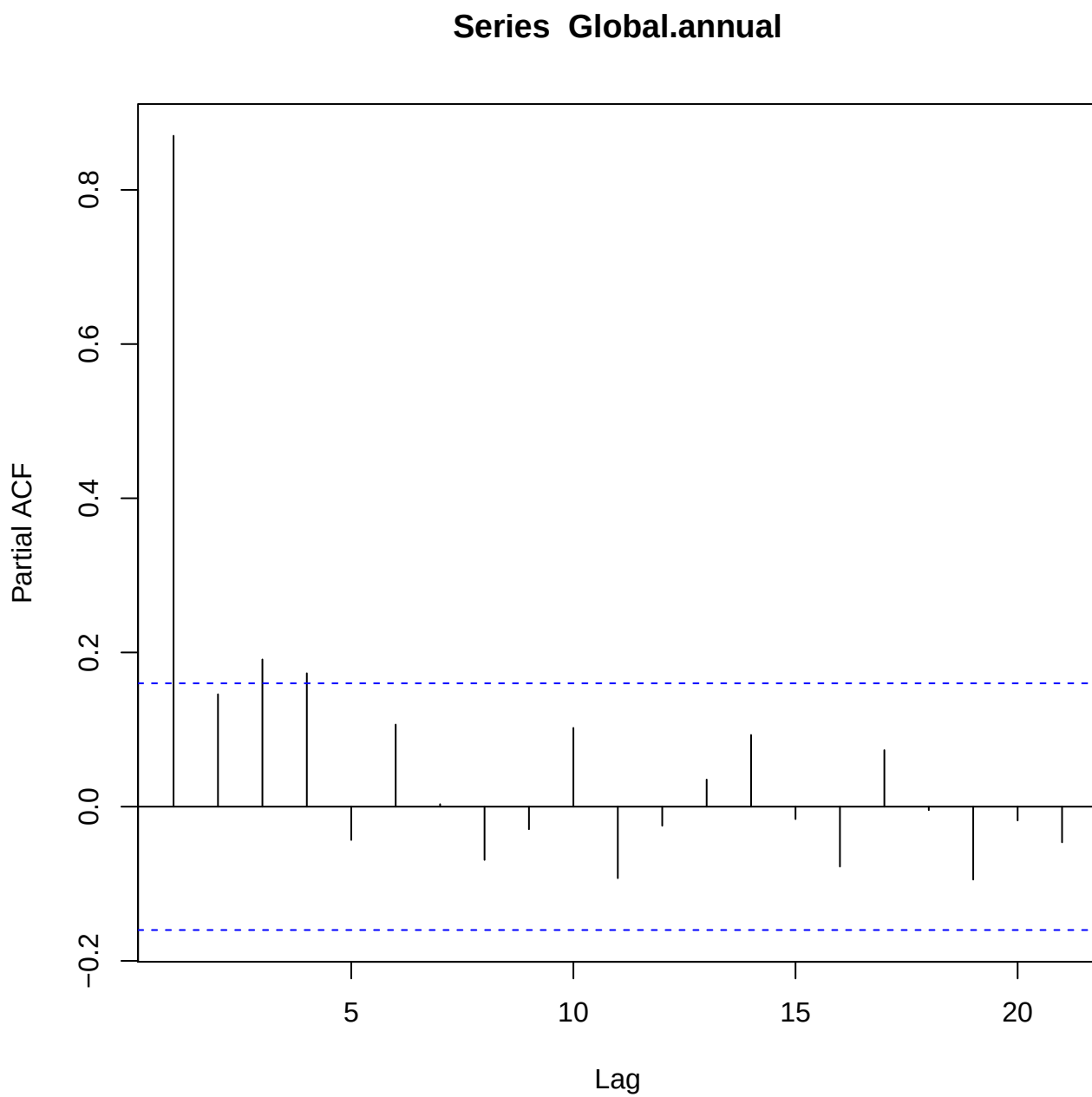
Coefficients:
      1      2      3      4
0.5876 0.0126 0.1112 0.2676

Order selected 4  sigma^2 estimated as  0.009865
```

This tells us that $\mu_x = -0.1382628 \approx -0.1383$ and that lag up to 4 periods behind are significant. Method “mle” indicates that we are asking **R** to use the Maximum Likelihood Method to do the estimation of the parameters.

11. Upon checking the `pacf` with `pacf(Global.annual.mean.adjusted)` or `pacf(Global.annual)`, we get the following partial correlogram in Figure 7. Since the most largest significant lag is the lag of order 4, that means all lags from order 1 to 4 are significant, even though some of those may appear to lie within the dotted blue lines in the partial correlogram in Figure 7.

Figure 7: Partial Correlogram of Annual Temperature Deviations



12. We can also check that the residuals of the autoregressive fit follow a pure white noise model. If this is so, then it means that we have completely fitted out autoregressive model to the required order of the lags that are significant. The correlogram of the residuals is shown in Figure 8 and we see that it has a white noise type correlogram profile.

```
> acf(Global.ar$res[-c(1,2,3,4)])
```

Note that we have to exclude the first 4 observations since they do not have residuals associated with them as they are used for the first 4 starting values in the fitting.

13. Since the residuals consist of pure white noise without any trends, this tells us that the supposedly increasing trend towards the end of the series isn't a significant trend at all and can be explained as a random outcome.
14. Putting everything together and rounding up the co-efficient estimates, our model is

$$y_t = 0.5876y_{t-1} + 0.0126y_{t-2} + 0.1112y_{t-3} + 0.2676y_{t-4} + w_t,$$

where $y_i = x_i - (-0.1383) = x_i + 0.1383$. Thus in terms of the original series, we have

$$\begin{aligned} x_t = & -0.1383 + 0.5876(x_{t-1} + 0.1383) + 0.0126(x_{t-2} + 0.1383) \\ & + 0.1112(x_{t-3} + 0.1383) + 0.2676(x_{t-4} + 0.1383) + w_t. \end{aligned}$$

15. Therefore if we want to predict ahead, say in the near term future for the 151st observation onwards, we can recursively use the prediction formula

$$\hat{x}_t = -0.1383 + 0.5876(x_{t-1} + 0.1383) + 0.0126(x_{t-2} + 0.1383) + 0.1112(x_{t-3} + 0.1383) + 0.2676(x_{t-4} + 0.1383),$$

for $t = 151, 152, \dots$. For example, if we are currently in 2005 and want to predict the mean annual global temperature deviation for 2006, which is $\hat{x}_{151}$, then

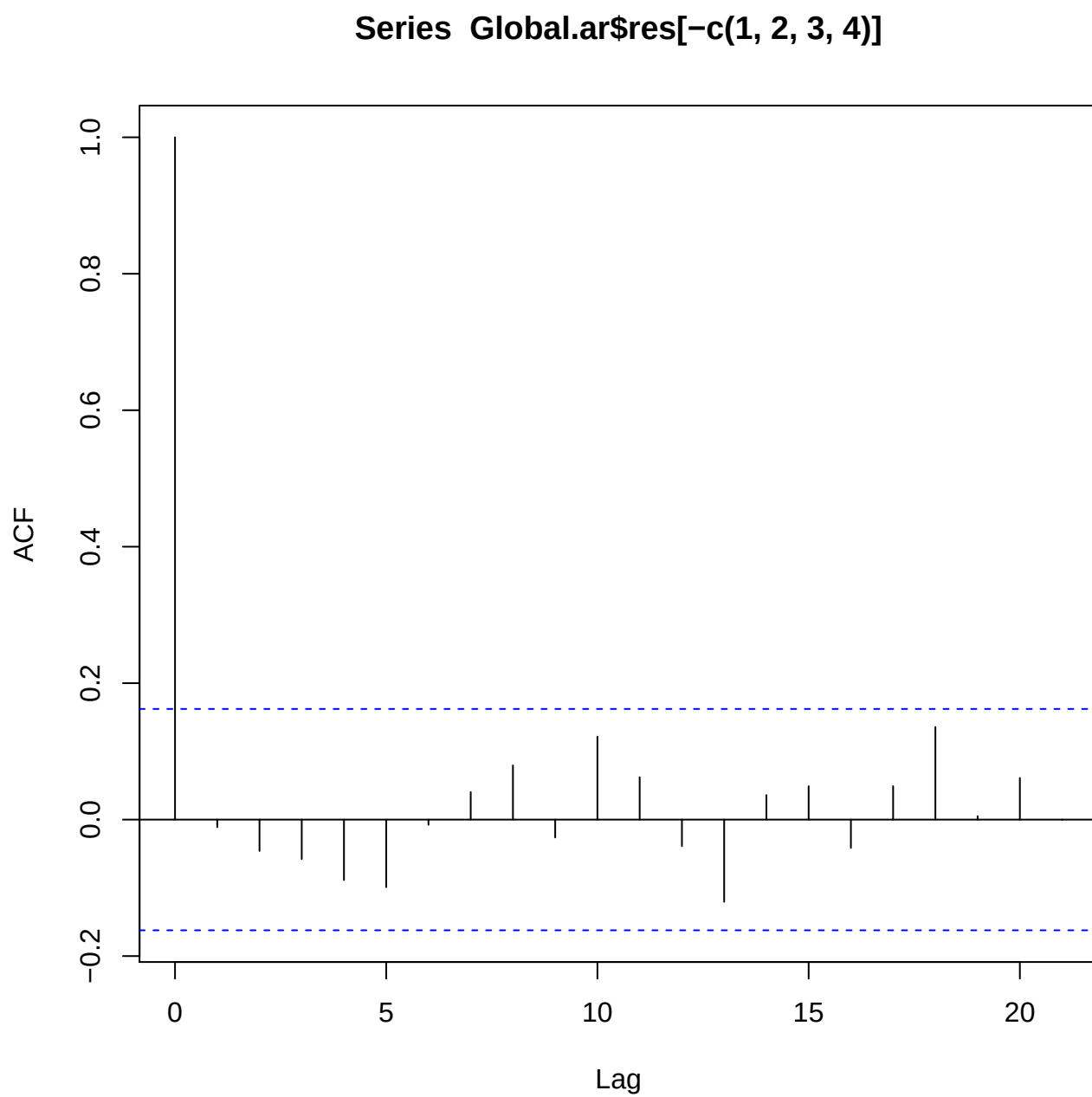
$$\hat{x}_{151} = -0.1383 + 0.5876(x_{150} + 0.1383) + 0.0126(x_{149} + 0.1383) + 0.1112(x_{148} + 0.1383) + 0.2676(x_{147} + 0.1383),$$

using the values of the last 4 years in the data, and

$$\hat{x}_{152} = -0.1383 + 0.5876(\hat{x}_{151} + 0.1383) + 0.0126(x_{150} + 0.1383) + 0.1112(x_{149} + 0.1383) + 0.2676(x_{148} + 0.1383),$$

to predict for 2007 and so on.

Figure 8: Correlogram of Residuals to Annual Temperature Autoregressive Fit



3 Moving Average Models

1. A *Moving Average* (MA) process of order q is a linear combination of the current white noise term and the q most recent past white noise terms and is defined by

$$x_t = w_t + \beta_1 w_{t-1} + \beta_2 w_{t-2} + \cdots + \beta_q w_{t-q},$$

where $\{w_t\}$ is white noise with zero mean and variance σ_w^2 .

2. Using the backward shift operator B notation, this can be expressed as

$$x_t = (1 + \beta_1 B + \beta_2 B^2 + \cdots + \beta_q B^q) w_t = \phi_q(B) w_t,$$

where $\phi_q(B)$ is a polynomial of B of order q .

3. The $MA(q)$ process x_t has zero mean because all the w_t have zero means and x_t is a weighted sum of $w_t, w_{t-1}, \dots, w_{t-q}$.
4. The variance is the sum of the variance of each individual terms in the right hand side of the equation defining x_t in 3.1, and thus it is

$$\begin{aligned} & \sigma_w^2 + \beta_1^2 \sigma_w^2 + \beta_2^2 \sigma_w^2 + \cdots + \beta_q^2 \sigma_w^2 \\ &= \sigma_w^2 (1 + \beta_1^2 + \beta_2^2 + \cdots + \beta_q^2), \end{aligned}$$

because each of the white noise terms are mutually independent and have the same variance σ_w^2 .

5. The autocorrelation function, for $k \geq 0$, is given by

$$\rho(k) = \begin{cases} 1 & \text{for } k = 0 \\ \frac{\sum_{i=0}^{q-k} \beta_i \beta_{i+k}}{\sum_{i=0}^q \beta_i^2} & \text{for } k = 1, \dots, q \\ 0 & \text{for } k > q, \end{cases}$$

where $\beta_0 = 1$.

6. The moving average $MA(q)$ process can be expressed as a stationary autoregressive process of infinite order without an error term if the absolute value of the roots of $\phi_q(B)$ are all greater than 1 in magnitude.

7. **Example 3.** We simulate an $MA(2)$ process

$$x_t = w_t + 0.5w_{t-1} + 0.3w_{t-2}.$$

The **R** command

```
> polyroot(c(1,0.5,0.3))  
[1] -0.833333+1.624466i -0.833333-1.624466i
```

tells us that there are two roots which exceed 1 in absolute magnitude since

$$\sqrt{0.833333^2 + 1.624466^2} > 1.$$

Now we do the simulation of 100 periods.

```
> set.seed(1)  
> b <- c(0.5,0.3)  
> x <- w <- rnorm(100)  
> for (t in 3:100){  
+ for (j in 1:2) x[t] <- x[t] + b[j]*w[t-j]  
+ }  
> plot(x, type="l")  
> acf(x)  
> pacf(x)
```

A plot of the simulated process is shown in Figure 8. The correlogram is shown in Figure 9. Note that the correlogram or partial correlogram of a moving average process is not so clear cut compared to that of a pure autoregressive process, for example, compared to that in Figure 7 of the Global Annual Temperature Deviations. The ACF or the PACF of an autoregressive process decays much more rapidly beyond that of the last lag order of significance (exponential type decay), whereas for a moving average process, due to random sampling variations, some of the correlations of the lags further down may also be marginally significant.

Figure 9: A Simulated $MA(2)$ Process

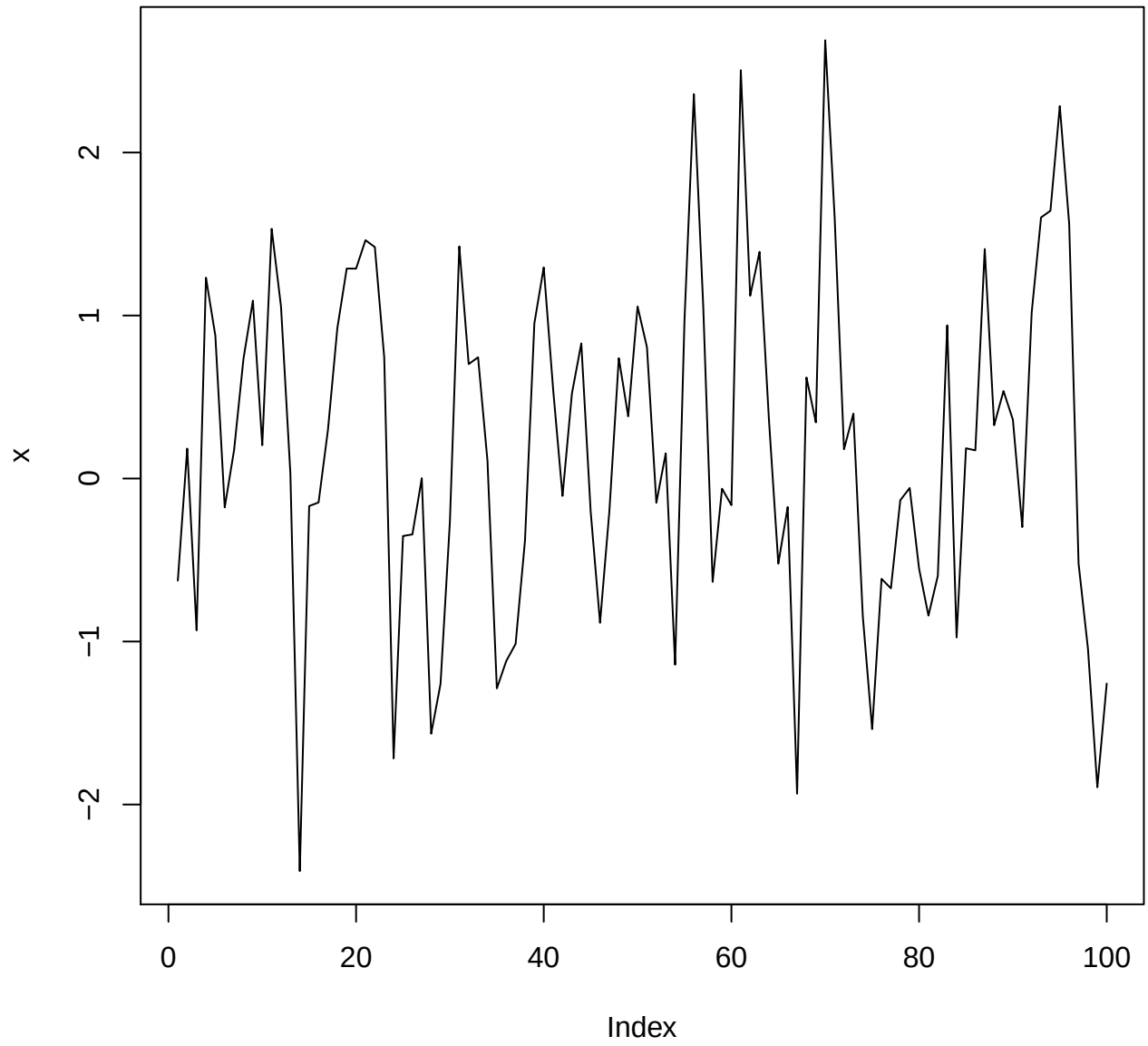
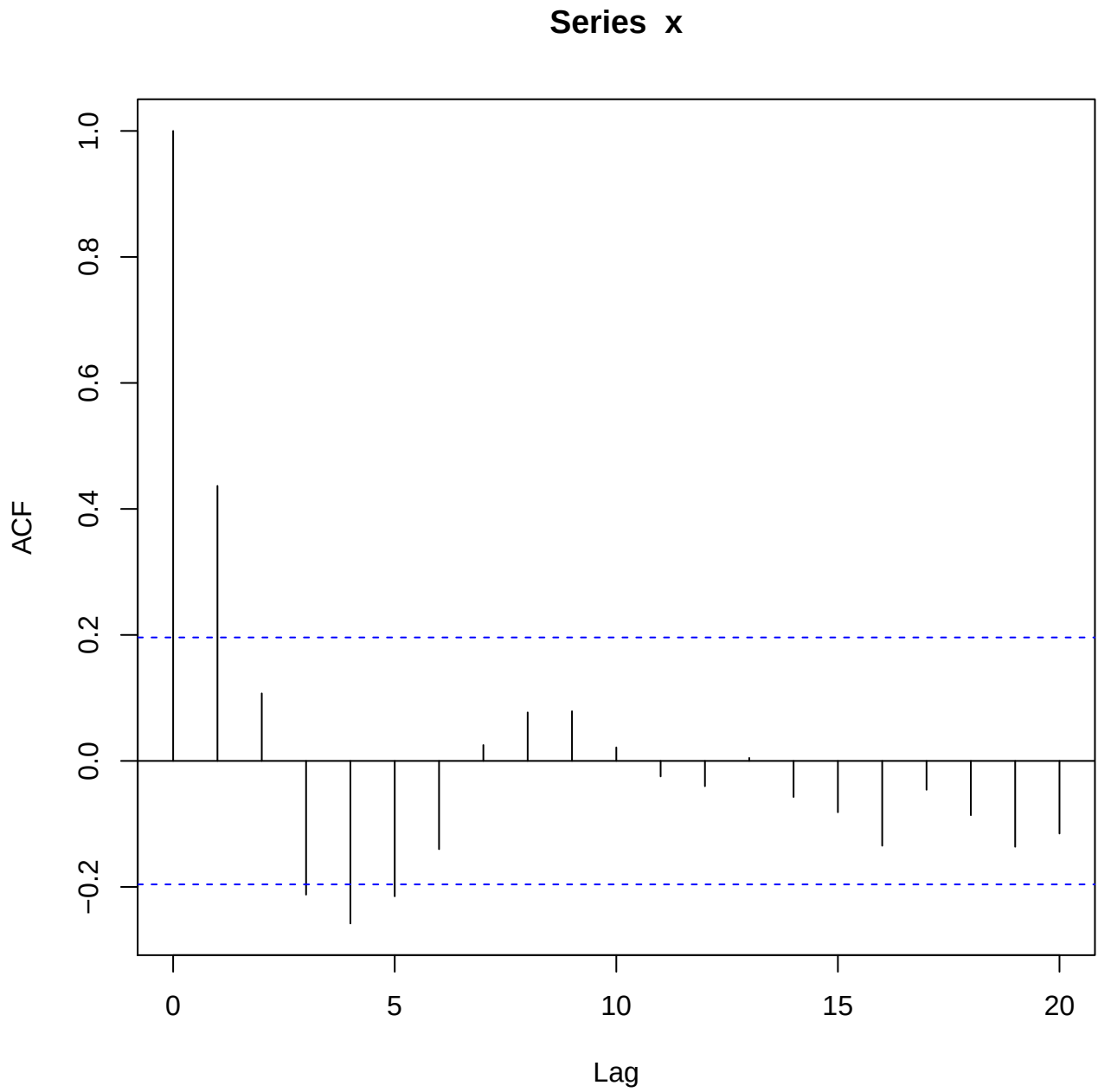


Figure 10: Correlogram of Simulated $MA(2)$ Process



8. **Example 4.** Pounds to NZD exchange rate. We read in the data from the pounds_nz.dat file.

```
> x <- read.table("D:/MATH 5045/Data/pounds_nz.dat", header=T)
```

The data is actually the average exchange rate over each quarter starting from the first quarter 1991, so we convert it to a time series.

```
> x.ts <- ts(x, st=1991, fr=4)
```

You can do a plot yourself to see how it looks like. The partial correlogram of the series shows that only the first order lag is significant.

We will try fitting both an $AR(1)$ and an $MA(1)$ separately to the data to see what happens. We use the command `arima(data, order=(p,d,q))` where p is the order of the autoregressive component, d is the difference (will be needed when there is seasonality or periodicity) and q is the order the moving average component. If we are fitting only a purely autoregressive model, then d and q are set to zero or alternatively use the `ar( )` command. If we are fitting only a purely moving average model, then set p and d to zero.

We fit the $AR(1)$ model first.

```
> x.ar <- arima(x.ts, order=c(1,0,0))
> x.ar
```

```
Call:
arima(x = x.ts, order = c(1, 0, 0))
```

Coefficients:

	ar1	intercept
	0.9439	3.0111
s.e.	0.0458	0.2953

```
sigma^2 estimated as 0.01818: log likelihood = 21.7, aic = -37.4
```

We look at the ACF of the residual of the fit.

```
> acf(x.ar$resid[-1])
```

We see that it does not look like a white noise ACF since the correlation at lag 1 is significant.

Now we try fitting the $MA(1)$ model and also look at the ACF of its residuals.

```
> x.ma <- arima(x.ts, order=c(0,0,1))
> x.ma
```

```
Call:
arima(x = x.ts, order = c(0, 0, 1))
```

```
Coefficients:
          ma1  intercept
          1.000      2.8329
s.e.    0.072      0.0646
```

```
sigma^2 estimated as 0.04172:  log likelihood = 4.76,  aic = -3.53
> acf(x.ma$resid[-1])
```

Again this does not look like the ACF of a white noise process. Thus in both cases, it looks like the neither of the models fully explain the data, since otherwise the residuals will be pure white noise.

We will see in the next section that we can combine the two processes into the autoregressive moving average (ARMA) model and this will give us a better fit.

4 ARMA Models

1. When necessary, we can combine both an autoregressive process and a moving average process to do the modelling. This is called an *autoregressive moving average* process or ARMA process for short.
2. A time series $\{x_t\}$ follows an $ARMA(p, q)$ process if

$$x_t = \alpha_1 x_{t-1} + \alpha_2 x_{t-2} \cdots + \alpha_p x_{t-p} \\ + w_t + \beta_1 w_{t-1} + \beta_2 w_{t-2} + \cdots + \beta_q w_{t-q}.$$

3. Using the backward shift operator notation, the $ARMA(p, q)$ process in item 4.2 can be written as

$$\theta_p(B)x_t = \phi_q(B)w_t.$$

4. The following points should be noted about an $ARMA(p, q)$ process.
 - (a) The process is stationary when the roots of θ are greater than one in absolute value.
 - (b) The process is invertible when the roots of ϕ are greater than one in absolute value.
 - (c) An $AR(p)$ model is a special case $ARMA(p, 0)$.
 - (d) An $MA(q)$ model is a special case $ARMA(0, q)$.
 - (e) When fitting to data, an ARMA model will often require less number of parameters than a single MA or AR model.
 - (f) When θ and ϕ share a common factor, a stationary model can be simplified, for example,

$$(1 - \frac{1}{2}B)(1 - \frac{1}{3}B)x_t = (1 - \frac{1}{2}B)w_t$$

can be simplified to $(1 - \frac{1}{3}B)x_t = w_t$.

5. **Example 4** continued. Now we fit an $ARMA(1, 1)$ model to the Pounds–NZD data.

```
> x.arma <- arima(x.ts, order=c(1,0,1))
> x.arma
```

```
Call:
arima(x = x.ts, order = c(1, 0, 1))
```

```
Coefficients:
          ar1          ma1    intercept
      0.8925   0.5319       2.9597
s.e.  0.0759   0.2021       0.2435
```

```
sigma^2 estimated as 0.01505:  log likelihood = 25.14,  aic = -42.27
```

We also check the correlogram plot of the residuals after fitting the *ARMA*(1,1) model

```
> acf(x.arma$resid)
```

and it resembles that of a white noise process. Thus we conclude that the *ARMA*(1,1) model provides the best fit among the three models. The equation for this model is

$$x_t - 2.9597 = +0.8925(x_{t-1} - 2.9597) + w_t + 0.5319w_{t-1}.$$

We can also obtain further information from the AIC (Akaike Information Criterion) numbers of the 3 models. For the *AR*(1), it is -37.4 , for the *MA*(1), it is -3.53 and for the *ARMA*(1,1), it is -42.27 . The AIC measures the trade-off between the number of parameters and the fit of the model. Generally the smaller the AIC, the better the fit with the optimum number of parameters. For the *ARMA*(1,1), the happens to be the smallest (the most negative in this case).